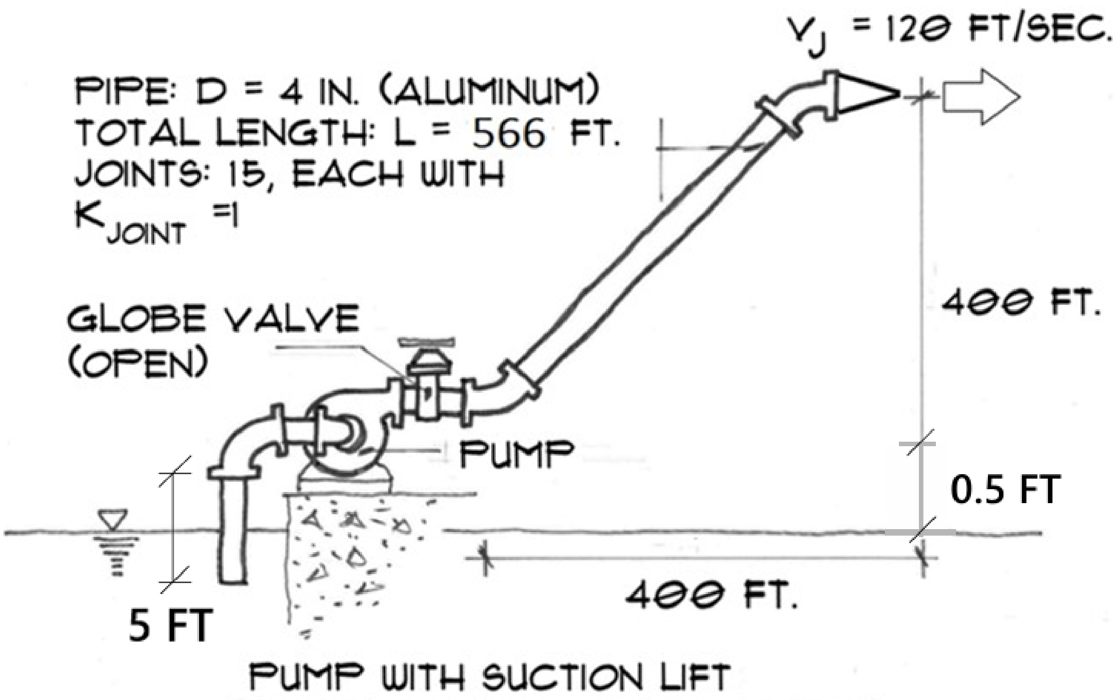
Question 1 (3.2)

Cooling water is pumped from a reservoir to equipment at a construction site through a 4-inch inner diameter aluminum pipe system, as shown below. The discharge pipe, 566 feet in length, is connected to a globe valve. Note that aluminum pipes have an average roughness similar to drawn tubing. The discharge pipe also includes 15 joints (each with a loss coefficient Kjoint=1, though not shown in the figure). The required flow rate is 600 gpm, and the water must exit the spray nozzle (a component with a reduced cross-sectional area, K=14) at 120 fps. The suction pipe is a short 5-foot section (indicated in the figure), with the pump positioned 0.5 feet above the water surface. Please address the following:

(1) *Estimate the required motor power considering a pump efficiency of 80%.*

(2) *Assess whether cavitation will occur if the NPSHR for this pump is 20 ft.*



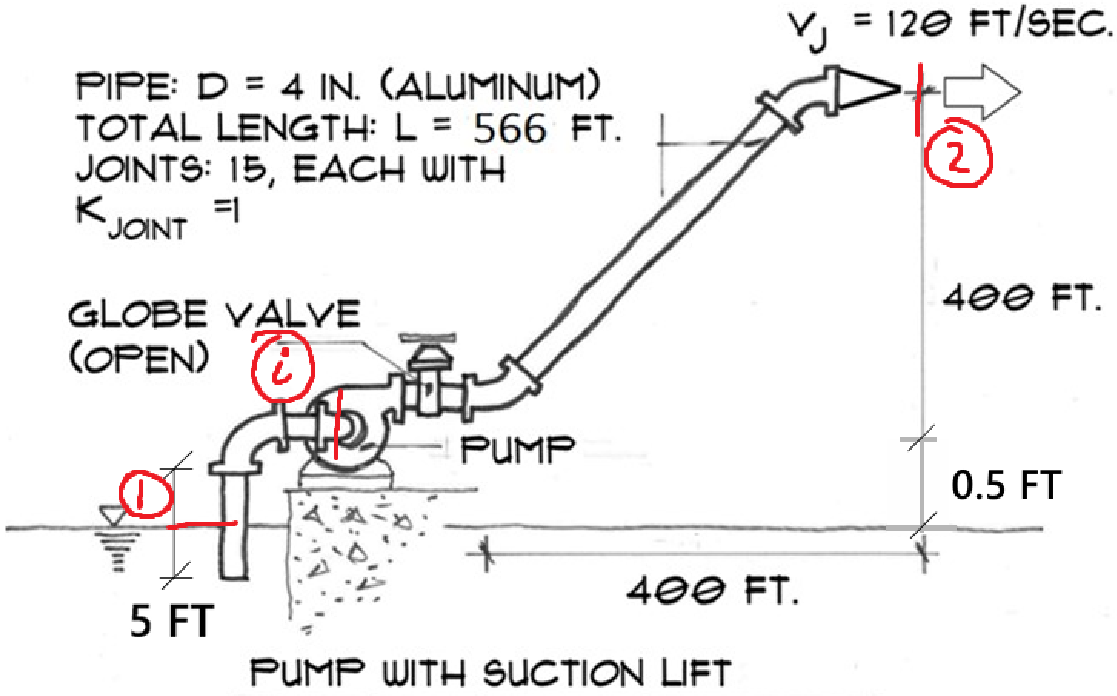
Note: the indicated lengths are not to scale.

1 Assumptions

1. All pipe fittings are flanged
2. Water is room temperature, 70°F
3. Two 45° bends and two 90° bends are used

4. The reservoir is sufficiently large that the water velocity is negligible

2 Sketches



Note: the indicated lengths are not to scale.

3 Analysis

3.1 Pipe Friction

First the properties of water can be found from Table A-9E at 70°F:

$$\mu = 6.556 \times 10^{-4} \text{ lbm/ft-s} \quad \text{and} \quad \rho = 62.30 \text{ lbm/ft}^3$$
$$P_{\text{vapour}} = 0.3632 \text{ psia}$$

The Reynolds number can be calculated as:

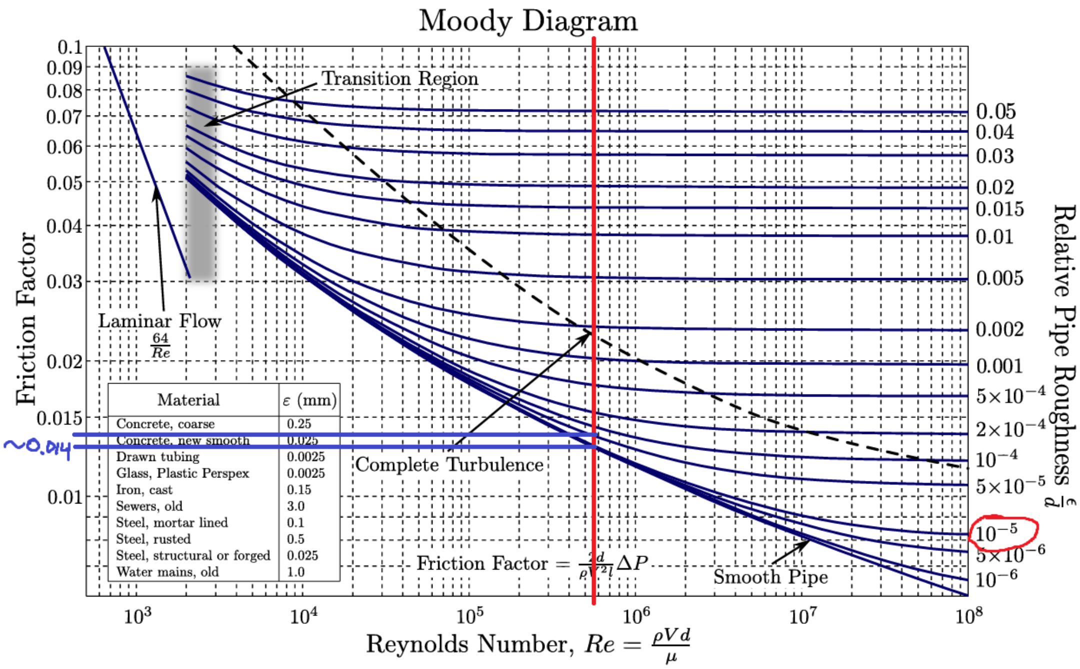
$$\text{Re} = \frac{\rho v D}{\mu} = \frac{4\rho\dot{V}}{\pi\mu D} = \frac{4 \cdot 62.30 \cdot 600\text{gpm}}{\pi \cdot 6.556 \times 10^{-4} \cdot \frac{4}{12} \text{ ft}} \cdot \frac{1 \text{ ft}^3}{7.48 \text{ gallons}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 4.85 \times 10^5$$

This flow is turbulent.

The relative roughness is

$$\frac{\varepsilon}{D} = \frac{0.000005 \text{ ft}}{\frac{4}{12} \text{ ft}} = 0.000015 = 1.5 \times 10^{-5}$$

From the Moody diagram, the friction factor is $f = 0.014$.



4 Outlet Reynolds

We know the outlet velocity is 120 fps, and that

$$\dot{V} = \pi \frac{D^2}{4} v$$

$$\implies D = \sqrt{\frac{4\dot{V}}{\pi v}} = \sqrt{\frac{4 \cdot 600\text{gpm}}{\pi \cdot 120\text{fps}}} \cdot \frac{1 \text{ ft}^3}{7.48 \text{ gallons}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 0.119 \text{ ft}$$

Therefore the Reynolds number at the outlet is

$$\text{Re} = \frac{4\rho v}{\pi \mu D} = \frac{4 \cdot 62.30 \cdot 120}{\pi \cdot 6.556 \times 10^{-4} \cdot 0.119} = 1.36 \times 10^6$$

This flow is also turbulent, $\alpha_2 = 1.06$.

5 Energy Equation

The energy equation can be written as

$$
\begin{aligned}
h_{\text{pump}} &= \left(\frac{P_2}{\rho g} + \alpha_2 \frac{v_2^2}{2g} + z_2\right) - \left(\frac{P_1}{\rho g} + \cancelto{0}{\alpha_1 \frac{v_1^2}{2g}} + \cancelto{0}{z_1}\right) + H_{\text{lT}} \\
&= \cancelto{0}{\frac{P_2 - P_1}{\rho g}} + \alpha_2 \frac{v_2^2}{2g} + z_2 + H_{\text{lT}} \\
&= \alpha_2 \frac{v_2^2}{2g} + z_2 + H_{\text{lT}}
\end{aligned}
$$

Where

$$
\begin{gathered}
H_{\text{lT}} = \\
frac v^2_{\text{pump}} 2g \left(f \frac{L}{D} + \sum K_L\right)
\end{gathered}
$$

From Table A.14 for losses,

$$
\begin{aligned}
K_{\text{globe}} &= 6 \quad K_{\text{exit}} = 1 \\
K_{45^\circ} &= 0.19 \quad K_{\text{nozzle}} = 14 \\
K_{\text{inlet}} &= 0.78 \quad K_{\text{joint}} = 1 \\
K_{90^\circ} &= 0.3
\end{aligned}
$$

Then,

$$
\begin{aligned}
\sum K_L &= K_{\text{globe}} + K_{\text{exit}} + 2K_{45^\circ} + K_{\text{nozzle}} + K_{\text{inlet}} + 15K_{\text{joint}} + 2K_{90^\circ} \\
&= 6 + 1 + 2(0.19) + 14 + 0.78 + 15 + 2(0.3) = 37.76
\end{aligned}
$$

Then,

$$
\begin{aligned}
H_{\text{lt}} &= \frac{8Q^2}{\pi^2 g D^4}\left(f \frac{L}{D} + \sum K_L\right) \\
&= \frac{8 \cdot 600^2}{\pi^2 \cdot 32.2 \cdot 0.119^4} \cdot \left(\frac{1 \text{ ft}}{7.48 \text{ gal}} \cdot \frac{1 \text{ min}}{60 \text{ s}}\right)^2 \cdot \left(0.014 \cdot \frac{566}{4/12} + 37.76\right) \\
&= 226 \text{ ft}
\end{aligned}
$$

and

$$
\begin{aligned}
h_{\text{pump}} &= \alpha_2 \frac{v_2^2}{2g} + z_2 + H_{\text{lT}} \\
&= 1.06 \cdot \frac{120^2}{2 \cdot 32.2} + 400 + 226 = 863 \text{ ft}
\end{aligned}
$$

5.1 Power

The power required for $\eta = 0.80$ is

$$\boxed{\begin{aligned}\text{bhp} &= \frac{\rho \dot{V} h_{\text{pump}}}{\eta} \\ &= \frac{62.30 \cdot 600 \cdot 863}{0.80} \cdot \frac{1\ \text{ft}^3}{7.48\ \text{gallons}} \cdot \frac{1\ \text{min}}{60\ \text{s}} = 163\ \text{hp}\end{aligned}}$$

5.2 Cavitation

NPSH is defined as

$$\text{NPSH} = \left(\frac{P}{\rho g} + \frac{v_{\text{avg, in}}^2}{2g}\right)_{\text{pump, in}} - \frac{P_{\text{vapour}}}{\rho g}$$

We need to determine the pressure at the pump inlet. By energy balance,

$$\left(\frac{P_i}{\rho g} + \alpha_i \frac{v_i^2}{2g} + z_i\right) + H_{\text{lT}} = \left(\frac{P_1}{\rho g} + \alpha_1 \frac{v_1^2}{2g} + z_1\right)$$

Assuming $v_1 = v_i$ and defining $z_1 = 0$,

$$\frac{P_i}{\rho g} = \frac{P_{\text{atm}}}{\rho g} - z_i - H_{\text{lT}}$$

H_{lT} is

$$H_{\text{lT}} = \frac{v_i^2}{2g}\left(f\frac{L}{D} + \sum K_L\right)$$

where,

$$\begin{aligned}\sum K_L &= K_{\text{inlet}} + 2K_{90^\circ} \\ &= 0.78 + 2(0.3) = 1.38\end{aligned}$$

and

$$\begin{aligned}v_1 = v_i &= \frac{\dot{V}}{\pi D^2} \\ &= \frac{4 \cdot 600}{\pi \cdot (4/12)^2} \cdot \frac{1\ \text{ft}^3}{7.48\ \text{gallons}} \cdot \frac{1\ \text{min}}{60\ \text{s}} \\ &= 15.31\ \text{fps}\end{aligned}$$

Then,

$$H_{\mathrm{lT}} = \frac{15.31^2}{2 \cdot 32.2}\left(0.014 \cdot \frac{5}{4/12} + 1.38\right) = 5.79 \text{ ft}$$

Then NPSH,

$$\begin{aligned}
\mathrm{NPSH} &= \frac{P_{\mathrm{atm}} - P_{\mathrm{vapour}}}{\rho g} - z_i - H_{\mathrm{lT}} \\
&= \frac{14.7 - 0.3632}{62.30 \cdot 32.2} \cdot \frac{32.2\mathrm{lb\ ft}^2/\mathrm{s}}{1\mathrm{lbf}} \cdot \left(\frac{12\mathrm{in}}{1\mathrm{ft}}\right)^2 - 0.5 - 5.79 \\
&= 26.8 \text{ ft}
\end{aligned}$$

Since NPSH is greater than NPSHR, cavitation will not occur.

6 Conclusion

Cavitation will not occure in the pump and the required motor power is 163 hp assuming an efficiency of 80%. A strainer or foot valve would be a good addition tothe inlet suction pipe. The fluid velocity is also quite high, so a larger pipe diameter may be considered to reduce the velocity and pressure drop.

Question 2 (3.8)

A farmer is considering a pipe-pump assembly to irrigate a field during the summer. The water source is an underground well, 250 feet deep, with the highest free water surface approximately 8 feet below grade. To minimize pump operation time, the farmer plans to fill a 2,500-gallon tank within 4 hours, providing adequate water for farm operations. The vented cylindrical tank must be no more than 10 feet in length and should include valved piping connections. The tank can be placed at any suitable location within the system. Given the field and well's location, the pumps must be positioned no more than 600 feet from the well. The pipe will connect to an existing irrigation system, which is 750 feet from the well and has a 10-foot length. The terrain from the well to the field is flat. Please complete the following tasks:

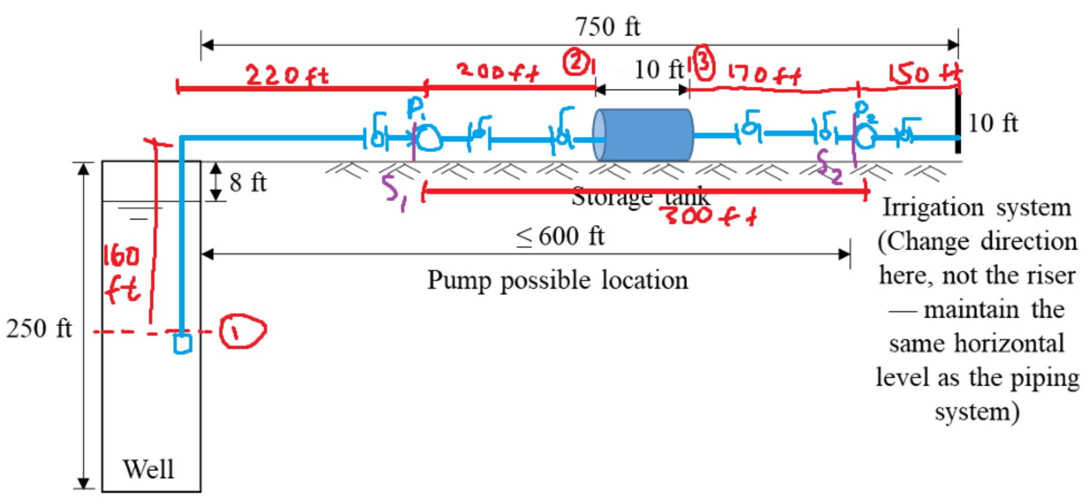
(1) *Design a constant cross-area piping system that includes 2 pumps. Specify the pipe size, pipe length, pipe material, pump sizes, and necessary piping accessories to meet the farmer's water requirements. For pump selection, you may refer to Taco pumps: http://www.tacocomfort.com/produc category/pumps/. Performance curves for the pumps can be found in the "Related Documents/Data Sheets" section for each series on the website. Other pump brands are also acceptable. Please attach the selected pump performance curve to your assignment.*

(2) *Specify the maximum depth of the pipe inlet within the well.*

1 Assumptions

- Temp is room temperature, 70°F
- Irrigation system is same size as connected pipe
- Pipe is A40 steel
- Use friction loss of ¡ 3 ft/100ft
- Constant pipe size
- Ball valves used at inlet
- All connections are threaded
- All bends 90°
- Foot valve used at inlet
- Constant pipe size
- Flow velocity is 5 fps, pump station line max, lower than erosion limit
- Sharp edge exit & entry

2 Sketch



3 Analysis: Pump Selection

3.1 Pipe Sizing

The flow rate can be calculated as:

$$\begin{aligned} Q &= \frac{2500 \text{ gallons}}{4 \text{ hours}} \cdot \frac{1 \text{ hour}}{60 \text{ min}} \\ &= 10.42 \text{ gpm} \\ &= 10.42 \cdot \frac{1 \text{ ft}^3}{7.48 \text{ gallons}} \cdot \frac{1 \text{ min}}{60 \text{ s}} \\ &= 0.0232 \text{ ft}^3/\text{s} \end{aligned}$$

From pipe diameter can be determined since the pipe length is sketched to be 10ft:

$$\begin{aligned} V_{\text{cylinder}} &= \frac{\pi D^2}{4} \cdot L \\ \implies D &= \sqrt{\frac{4V_{\text{cylinder}}}{\pi L}} \\ &= \sqrt{\frac{4 \cdot 2500 \text{ gallons}}{\pi \cdot 10 \text{ ft}}} \cdot \frac{1 \text{ ft}^3}{7.48 \text{ gallons}} \\ &= 2.53 \text{ ft} \end{aligned}$$

On chart in Figure A.4, a pipe of 1.5 inch diameter will result in a water velocity of 1.8 fps, friction loss of 1.5 ft/100ft. A threaded connection is applicable for this use.

3.2 Reynolds Number

The Reynolds number for the pipe can be calculated as:

$$\begin{aligned} \text{Re} &= \frac{\rho v D}{\mu} = \frac{62.3 \times 1.8 \times 1.5/12}{6.556 \times 10^{-4}} \\ &= 2.1 \times 10^5 \end{aligned}$$

This flow is turbulent.

3.3 Energy Balance

For pump 1:

$$h_{1-p_1} = \frac{1.5}{100}(160 + 220 + 200) = 8.7 \text{ ft}$$

and pump 2:

$$h_{1-p_2} = \frac{1.5}{100}(170 + 150 + 10) = 4.95 \text{ ft}$$

Minor losses:

$$\begin{aligned}
k_{\text{foot valve}} &= 1.25 \\
k_{\text{exit}} &= 1 \\
k_{\text{entrance}} &= 0.5 \\
k_{\text{elbow }90^\circ} &= 1.225 \\
k_{\text{ball valve}} &= 0.05
\end{aligned}$$

for pump 1:

$$\begin{aligned}
h_{lm} &= \frac{v^2}{2g}\sum k_L \\
&= \frac{v^2}{2g}\left(k_{\text{foot valve}} + 2k_{\text{elbow }90^\circ} + 3k_{\text{ball valve}} + k_{\text{exit}}\right) \\
&= \frac{1.8^2}{2\cdot 32.2}\left(1.25 + 2\cdot 1.225 + 3\cdot 0.05 + 1\right) \\
&= 0.244 \text{ ft}
\end{aligned}$$

and for pump 2:

$$\begin{aligned}
h_{lm} &= \frac{v^2}{2g}\sum k_L \\
&= \frac{v^2}{2g}\left(k_{\text{entracnce}} + 2k_{\text{elbow }90^\circ} + 3k_{\text{ball valve}} + k_{\text{exit}}\right) \\
&= \frac{1.8^2}{2\cdot 32.2}\left(0.5 + 2\cdot 1.225 + 3\cdot 0.05 + 1\right) \\
&= 0.206 \text{ ft}
\end{aligned}$$

total head loss:

$$\begin{aligned}
h_{lt,p_1} &= 8.7 + 0.244 = 8.94 \text{ ft} \\
h_{lt,p_2} &= 4.95 + 0.206 = 5.16 \text{ ft}
\end{aligned}$$

Assume P_1 is full tank. $P_2 = P_{\text{atm}+\rho g d} = P_{\text{atm}+\rho g(z_2-z_1-x)}$. Also set $v_1 = v_2$ since constant diameter pipe. Letting z_1 be 0,

$$\begin{aligned}
h_{p_1} &= \left(\frac{P_{\text{atm}} + \rho g D}{\rho g} + \alpha\frac{v^2}{2g} + z_2\right) - \left(\frac{P_{\text{atm}} + \rho g(z_2 - z_1 - 8)}{\rho g} + \alpha\frac{v^2}{2g} + z_1\right) + h_{lt,p_1} \\
&= D + x + h_{lt,p_1}
\end{aligned}$$

Let $x = 8$ ft, and $x = 160$ ft in worse case

$$\begin{aligned}
h_{p_1} &= 6.53 + 8 + 8.94 = 23.47 \text{ ft} \\
h_{p_1} &= 6.53 + 160 + 8.94 = 171.7 \text{ ft}
\end{aligned}$$

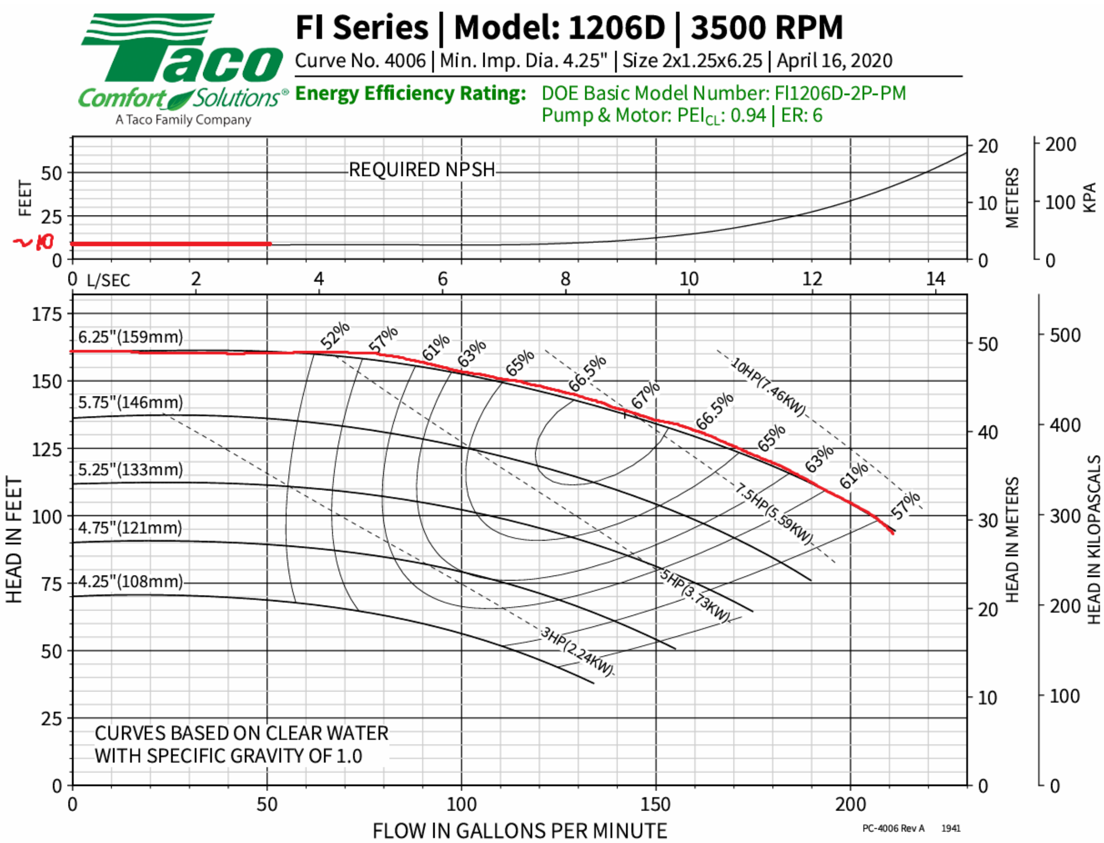
For pump 2, assume $P_3 = P_{\text{atm}} = P_4$, $v_3 = v_4$. $z_3 = z_4$. Then,

$$h_{p_2} = \left(\frac{P_{\text{atm}} + \rho g D}{\rho g} + \alpha\frac{v^2}{2g} + z_2\right) - \left(\frac{P_{\text{atm}}}{\rho g} + \alpha\frac{v^2}{2g} + z_1\right) + h_{lt,p_2}$$
$$h_{p_2} = h_{lt,p_2} = 5.16 \text{ ft}$$

3.4 Pump Requirements

For pump 1:

- $\sim$ 11 gpm
- 140 FT
- large pump head
- mounted is mosts suitable



This pump is suitable for the application,

- Casing size: 2” x 1.25 ” x 7.25”
- Impeller D: 6.25”
- Motor: 10 hp (oversized)

- Motor speed: 3500 rpm
- NPSHR: 10 ft

For pump 2:

Taco Comfort Solutions A Taco Group Company

Submittal Data Information

101-155

110 Series In-Line Circulators

EFFECTIVE: November 8, 2014 SUPERSEDES: February 1, 2013

JOB ____________ ENGINEER ____________ CONTRACTOR ____________ REP. ____________

ITEM	MODEL NO.	IMP. DIA.	G.P.M.	HEAD (FT.)	H.P.	ELEC. CHAR.

110 Series (110, 111, 112, 113, 120)
Red Baron Circulators

Rugged Motor Built to Last:
- Resilient mounted, split phase motor with built-in overload protector
- Equipped with sleeve bearings for quiet operation

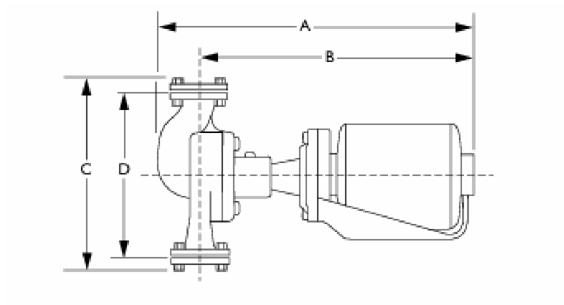
Proven Pump Construction:
- One piece non-ferrous impeller
- Stainless steel shaft
- Rugged bronze sleeve bearing
- Two piece carbon/ceramic seal assembly
- Durable one piece spring coupling

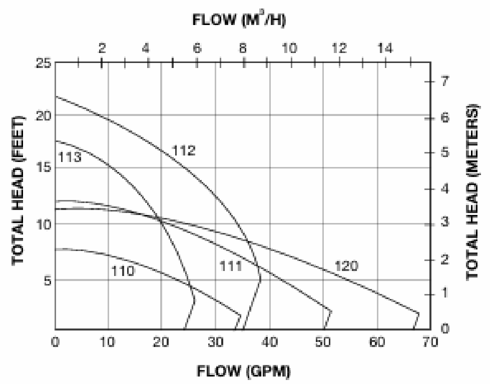
Factory Tested:
- Every circulator is operated before shipment

Easy, Low-Cost Service:
- No special tools required
- Nationwide parts availability

Accessories and Options:
- Available in all stainless steel construction
- Companion Freedom® flanges available in 3/4", 1", 1 1/4" and 1 1/2"
- Freedom® Shut-off flanges available in 3/4", 1", 1 1/4" and 1 1/2" allow easy isolation and service

NSF STAINLESS STEEL MODELS ONLY





Specifications:

MODEL NUMBER (1)	FLANGE SIZE	MOTOR (2) (115V/60 HZ/1 PH)		MAXIMUM TEMP. (3)	PRESSURE RATING	DIMENSIONS								APPROX. SHIPPING	
						A		B		C		D			
		HP	RPM			IN.	MM	IN.	MM	IN.	MM	IN.	MM	LBS.	KG
110	3/4", 1", 1 1/4", 1 1/2"	1/12	1725	240°F	125 psi	14 5/8	371.5	12 5/8	320.7	7 7/8	200.0	6 5/16	160.3	21	9.5
111		1/8		240°F (3)		16 1/4	412.8	13 7/8	352.4	10 1/4	260.4	8 3/4	222.3	26	11.8
112		1/3	3450	240°F		16 1/2	419.1	14 1/2	368.3	7 7/8	200.0	6 3/8	161.9	28	12.7
113		1/8	1725	240°F (3)		16 1/4	412.8	14	355.6	10 1/8	257.2	8 1/2	215.9	27	12.2
120	2"	1/6				16 7/8	428.6	14 1/4	362.0	13 1/2	342.9	11	279.4	46	20.9

(1) When specifying all stainless steel construction, add letter "S" after model number (e.g. 110S).
(2) Motors are available with other electrical characteristics – consult your Taco representative.
(3) 240° Intermittent, 200° Continuous.

Taco Comfort Solutions A Taco Group Company
Taco, Inc., 1160 Cranston Street, Cranston, RI 02920 | Tel: (401) 942-8000 | FAX: (401) 942-2360
Taco (Canada), Ltd., 8450 Lawson Road, Suite #3, Milton, Ontario L9T 0J8 | Tel: (905) 564-9422 | FAX: (905) 564-9436
Visit our web site: www.TacoComfort.com | Printed in USA |

- Casing size: 1.5" x 1.5" x 6 $\frac{5}{16}$ "

- Motor hp: 1/12 hp
- Motor speed: 1725 rpm

4 Analysis: Maximum Depth

Maximum depth occures when cavitation will occure for P_1 because this is an open loop system.

$$\text{NPSHA} = \frac{P_s}{\rho g} + \frac{v_s^2}{2g} - \frac{P_{\text{vapour}}}{\rho g}$$

For pump 1, energy balance assuming $v_1 = v_{s1}$, $z_1 = 0$, $P_1 = P_{\text{atm}} + \rho g(z_{s_1} - 8)$:

$$\frac{P_1}{\rho g} + \alpha_1 \frac{v_1^2}{2g} + z_{s_1} = \frac{P_{s_1}}{\rho g} + \alpha_{s_1} \frac{v_{s_1}^2}{2g} + z_{s_1} + h_{lt,s_1}$$

Then,

$$\frac{P_{s_1}}{\rho g} = \frac{P_{\text{atm}}}{\rho g} - 8 - h_{lt,s_1}$$

where

$$\begin{aligned} h_{lt} &= h_{1,s_1} + \frac{v^2}{2g} \sum K_L \\ &= \frac{1.5}{100}(z_{s_1} + 220) + \frac{1.8^2}{2 \cdot 32.2}(1.25 + 2 \cdot 1.225 + 3 \cdot 0.05 + 1) \\ &= 0.015 z_{s_1} + 3.189 \end{aligned}$$

Then the NPSHA is

$$\begin{aligned} \text{NPSHA} &= \frac{P_{\text{atm}}}{\rho g} - 8 - 0.015 z_{s_1} - 3.189 + \frac{1.8^2}{2 \cdot 32.2} - \frac{P_{\text{vapour}}}{\rho g} \\ &= \frac{P_{\text{atm}}}{\rho g} - 8 - 0.015 z_{s_1} - 3.189 + \frac{1.8^2}{2 \cdot 32.2} - \frac{0.3632}{62.3 \cdot 32.2} \cdot \frac{32.2 \text{ ft}}{1 \text{ psia}} \cdot \left(\frac{12 \text{ in}}{1 \text{ ft}}\right) \\ &= 21.99 - 0.015 z_{s_1} \end{aligned}$$

Set NPSHA to NPSHR to find the maximum depth:

$$21.99 - 0.015 z_{s_1} = 10 \implies z_{s_1} = 800 \text{ ft}$$

A depth of 120 ft will not experience cavitation.

For pump 2, assume $P_3 = P_{\text{atm}}$, $v_3 = v_{s_2}$, $z_3 = z_{s_2}$. Similarly,

$$\frac{P_{s_2}}{\rho g} = \frac{P_{\text{atm}}}{\rho g} - h_{lt,s_2}$$

For loss,

$$\begin{aligned} h_{lt,s_2} &= h_{1,s_1} + \frac{v^2}{2g}\sum K_L \\ &= \frac{1.5}{100}(200) + \frac{1.8^2}{2(32.2)}(0.5 + 2\cdot 0.05 + 1.225) \\ &= 3.09 \end{aligned}$$

Then NPSHA is

$$\begin{aligned} \text{NPSHA} &= \frac{P_{\text{atm}}}{\rho g} + \frac{v_{s_2}^2}{2g} - h_{lt,s_2} - \frac{P_{\text{vapour}}}{\rho g} \\ &= \frac{14.7 - 0.3632}{62.3(32.2)} \cdot \frac{32.2\text{ ft}}{1\text{ psia}} \cdot \left(\frac{12\text{ in}}{1\text{ ft}}\right) + \frac{1.8^2}{2\cdot 32.2} - 3.09 \\ &= 33.47\text{ ft} \end{aligned}$$

Pumps of this size don't usually have NPSHR values, so assume NPSHR is 10 ft. Since the NPSHA = 33.47 ft ¿ NPSHR = 10 ft, cavitation will not occur.

5 Conclusion

- The system is designed with a size of 1.5" diameter A40 steel pipe.
- Large 10hp base-mounted pump.
- Pump suction is lower than 5 fps to avoid erosion.
- Pipe may need support anchors between pumps to prevent sagging.
- Maximum depth of the pipe inlet is 800 ft before cavitation will occur.
- Pumps were oversized to account for changing conditions.

Pump	Model	Type	Construction	Flow Rate (gpm)	Fluid	Head Loss (ft)	Motor Size (hp)	Pump Speed (rpm)
Pump 1	Taco, FI/CI, Model 1207D	Centrifugal base mounted	Cast iron 2" x 1.25" x 7.25"	11	Water	140	10	3500
Pump 2	Taco, 110 Series, Model 110	Centrifugal in-line mounted	Cast iron 1.5" x 1.5" x 6 $\frac{5}{16}$ "	11	Water	5	1/12	1725